

Mathematics for Machine Learning

Vazgen Mikayelyan



Linear Mappings

Definition

For vector spaces V, W , a mapping $T : V \rightarrow W$ is called a linear mapping (or vector space homomorphism/ linear transformation) if

$$T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y}), \quad \forall x, y \in V, a, b \in \mathbb{R}$$

or equivalently,

- $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$
- $T(a\mathbf{x}) = aT(\mathbf{x})$

Definition

Consider a transformation $T : V \rightarrow W$, where V, W can be arbitrary sets. Then T is called

- *injective, if for any $\mathbf{x}, \mathbf{y} \in V$ it follows that $T(\mathbf{x}) \neq T(\mathbf{y})$ if and only if $\mathbf{x} \neq \mathbf{y}$.*
- *surjective, if $T(V) = W$.*
- *bijective if it is injective and surjective.*

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- *surjective*, if $T(V) = W$.
- *bijective* if it is injective and surjective.

Definition

Let V, W be vector spaces.

- $T : V \rightarrow W$ is called **isomorphism**, if it is linear and bijective.
- $T : V \rightarrow V$ is called **endomorphism**, if T is linear.
- $T : V \rightarrow V$ is called **automorphism**, if T is linear and bijective.
- We define $id_V : V \rightarrow V$, $id_v(\mathbf{x}) = \mathbf{x}$ as the **identity mapping** in V .

Definition

Vector spaces V and W are called isomorphic if there exists an isomorphism $T : V \rightarrow W$ and we write $V \cong W$.

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Finite-dimensional vector spaces V and W are isomorphic if and only if $\dim(V) = \dim(W)$.

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Example

Show that $R^{m \times n} \cong R^{mn}$.

Matrix Representation of Linear Mappings

Consider a vector space V with an (unordered) basis $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$. Let $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ be an ordered basis, and $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_n]$ be a matrix whose columns are the vectors $\mathbf{b}_1, \dots, \mathbf{b}_n$.

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Definition

For any $\mathbf{x} \in V$ there is a unique representation

$$\mathbf{x} = \alpha_1 \mathbf{b}_1 + \dots + \alpha_n \mathbf{b}_n.$$

Then $\alpha_1, \dots, \alpha_n$ are the coordinates of $\mathbf{x}$ with respect to B , and the vector

$$[\mathbf{x}]_B = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \in \mathbb{R}^n$$

*is the **coordinate vector** of $\mathbf{x}$ with respect to the ordered basis B .*

Matrix Representation of Linear Mappings

Definition

Let V, W be vector spaces with corresponding bases $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ and $C = (\mathbf{c}_1, \dots, \mathbf{c}_m)$. Moreover, let $T : V \rightarrow W$ be a linear mapping with

$$T(\mathbf{b}_j) = \alpha_{1j}\mathbf{c}_1 + \dots + \alpha_{mj}\mathbf{c}_m = \sum_{i=1}^m \alpha_{ij}\mathbf{c}_i.$$

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Then, we call the $m \times n$ -matrix A_T whose elements are given by

$$A_T(i, j) = \alpha_{ij}$$

the **transformation matrix** of T (with respect to the ordered bases B of V and C of W).

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Note that

$$A_T = [[T(\mathbf{b}_1)]_C \dots [T(\mathbf{b}_n)]_C] \text{ and } [T(\mathbf{x})]_C = A_T[\mathbf{x}]_B.$$

Matrix Representation of Linear Mappings

Example

Write the transformation matrix of the linear mapping (homomorphism) $T : V \rightarrow W$ w.r.t. ordered bases $B = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ of V and $C = (\mathbf{c}_1, \dots, \mathbf{c}_5)$ of W satisfying

$$T(\mathbf{b}_1) = \mathbf{c}_1 - \mathbf{c}_3 + 3\mathbf{c}_5$$

$$T(\mathbf{b}_2) = \mathbf{c}_2 + 2\mathbf{c}_4 + 5\mathbf{c}_5$$

$$T(\mathbf{b}_3) = \mathbf{c}_1 - 3\mathbf{c}_2 + \mathbf{c}_3$$

Using the transformation matrix A_T find the coordinate vector $[T(\mathbf{x})]_C$,

given that $[\mathbf{x}]_B = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$.

Matrix Representation of Linear Mappings

Example

Let's consider three linear transformations of a set of vectors in $\mathbb{R}^2$ with the transformation matrices

$$A_1 = \begin{bmatrix} \cos(\frac{\pi}{4}) & -\sin(\frac{\pi}{4}) \\ \sin(\frac{\pi}{4}) & \cos(\frac{\pi}{4}) \end{bmatrix}, \quad A_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_3 = \frac{1}{2} \begin{bmatrix} 3 & -1 \\ 1 & -1 \end{bmatrix}.$$

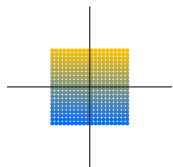
Matrix Representation of Linear Mappings

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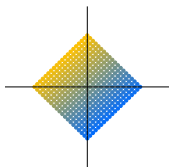
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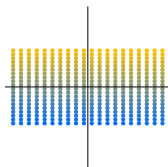
How are the vectors in figure a) transformed when applying each of the matrices A_i , $i = 1, 2, 3$.



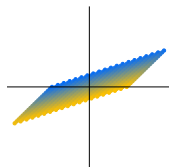
(a) Original data.



(b) Rotation by 45° .



(c) Stretch along the horizontal axis.



(d) General linear mapping.

Definition

For $T : V \rightarrow W$, we define the kernel/null space

$$\ker(T) := T^{-1}(\mathbf{0}_W) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}_W\}$$

and the image/range

$$\text{Im}(T) := T(V) = \{\mathbf{w} \in W : \exists \mathbf{v} \in V \text{ s.t. } T(\mathbf{v}) = \mathbf{w}\}.$$

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Proposition

- $T(\mathbf{0}_V) = \mathbf{0}_W$ and, therefore, $\mathbf{0}_V \in \ker(T)$.
- $\text{Im}(T) \subset W$ is a subspace of W , and $\ker(T) \subset V$ is a subspace of V .
- T is injective (one-to-one) if and only if $\ker(T) = \{0\}$.

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Let $A \in \mathbb{R}^{m \times n}$ be a matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, then

- $\text{span}[\mathbf{a}_1, \dots, \mathbf{a}_n] \subset \mathbb{R}^m$ is called **column space**,
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 $= \{x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n; x_1, \dots, x_n \in \mathbb{R}\} = \text{span}[\mathbf{a}_1, \dots, \mathbf{a}_n]$.

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- The kernel $\ker(T)$ is a subspace of $\mathbb{R}^n$.
- The kernel $\ker(T)$ is the null space of A .

Example

Find a basis for image and kernel of the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by

$$T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 - x_3 \\ x_2 + 3x_3 \end{bmatrix}$$

Image and Kernel

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Rank-Nullity Theorem

For vector spaces U, V and a linear mapping $T : U \rightarrow V$ it holds that

$$\dim(\ker(T)) + \dim(\operatorname{Im}(T)) = \dim(U),$$

equivalently for a matrix $A \in \mathbb{R}^{n \times m}$

$$\operatorname{nullity}(A) + \operatorname{rank}(A) = m,$$

where $\operatorname{nullity}(A)$ is the dimension of the null space $\operatorname{null}(A)$.

Definition

Given a vector space V , a **norm** on V is a function $\|\cdot\| : V \rightarrow [0, +\infty)$ with the following properties: For all $\mathbf{x}, \mathbf{y} \in V$, and all $\lambda \in \mathbb{R}$

- $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ (triangle inequality)
- $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$ (absolutely homogeneous)
- $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0$ iff $\mathbf{x} = 0$ (positive definite)

Norm

Example

The **Manhattan norm** on $\mathbb{R}^n$ is defined for $\mathbf{x} \in \mathbb{R}^n$ as

$$\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|$$

The Manhattan norm is also called ℓ_1 **norm**.

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Example

The length of a vector $\mathbf{x} \in \mathbb{R}^n$ is given by

$$\|\mathbf{x}\|_2 := \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{\mathbf{x}^T \mathbf{x}}.$$

This norm is called the **Euclidean norm**. The Euclidean norm is also called ℓ_2 **norm**.

Inner Products

Definition

A function $f : V \times V \rightarrow \mathbb{R}$ is called **bilinear** if

$$f(a\mathbf{x} + b\mathbf{y}, \mathbf{z}) = af(\mathbf{x}, \mathbf{z}) + bf(\mathbf{y}, \mathbf{z})$$

$$f(\mathbf{x}, a\mathbf{y} + b\mathbf{z}) = af(\mathbf{x}, \mathbf{y}) + bf(\mathbf{x}, \mathbf{z})$$

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Definition

A function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called **inner product on V** , if

- $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$ (symmetric)
- $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ iff $\mathbf{x} = 0$ (positive definite)
- $\langle \cdot, \cdot \rangle$ is a bilinear function

Example

The scalar product/dot product in $\mathbb{R}^n$, which is given by

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

The dot product is an inner product.

$\mathbb{R}^n$ with the dot product is called **Euclidean vector space**.

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Example

If we define for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$

$$\langle \mathbf{x}, \mathbf{y} \rangle := x_1 y_1 - (x_2 y_1 + x_1 y_2) + 2x_2 y_2$$

then $\langle \cdot, \cdot \rangle$ is an inner product but different from the dot product.